USB

F1
1A fast (466)
VBUS
VBUS_PROTECTED
SS14
D4
C22
100nF
D10
SMF5.0A
+5V
R40
100k
VBUS_F_ADC
R41
10k
C25
100nF
GND
+5V
VBUS
USB_DN
USB_DP
J1
USB-JST
MP
GND
TPD2S017
USB_ESP_DN
CH1_OUT
GND
CH1_IN
CH2_OUT
VCC
CH2_IN
USB_ESP_DP
USB_DP
+5V
D5
GND

ESP32-S3 MCU

+3V3
EN
BOOT
I2S_DOUT
I2S_WS
K5_SENSE_N
VBUS_F_ADC
LED_K
PTT_DRV
DOOR_DRV
MUTE_DRV
P4_SENSE_N
P5_SENSE_N
3V3
GND
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
PSRAM
USB_ESP_DN
USB_ESP_DP
I2C_SDA
I2C_SCL
I2S_MCLK
I2S_BCLK
I2S_DIN
PTT_SENSE_N
P4_ISO
U1
ESP32-S3-WROOM-1U-N16R8
C3
10uF
C6
100nF
+3V3
GND

POWER

U2
SGM2212-3.3
+5V
VIN
VOUT
GND
C2
10uF
C4
10uF
+3V3
GND

STATUS LED

D6
STATUS
LED_K
LED_A
R15
1k
+3V3
R27
1k
GND

BUS SWITCHES (PhotoMOS SSRs)

TALK (K1, dual NO SSR)
ch1: P2 <-> TALK_BRIDGE;
JP4 + R38/R39 200k precharges ch1
TALK_BRIDGE -> R28 2.2k -> TX_OUT;
ch2: TX_OUT <-> P3;
codec TX -> R26/C14 -> TALK_BRIDGE

K1
GAQW212GS
PTT_DRV
R4
220
K1A_A
R7
10k
R24
220
K1B_A
K1_LED_RET
P2
TALK_BRIDGE
TX_OUT
P3
R28
2.2kΩ
TX_PRECHARGE
JP3
R38
100k
PRECHG_IN
PRECHG_MID
TALK_BRIDGE
R39
100k
GND

DOOR OPENER (K2, NO SSR)
bridges P2<->P3 directly (dead short),
emulating the WF26 OT button (S1)

K2
GAQY212GS
DOOR_DRV
R8
10k
R25
10M
D11
1N4148W
R17
100k
R5
220
K2_A
WD_GATE
C20
2.2uF
DELAY_GATE
Q4
AO3400A
K2_K
Q3
AO3400A
C18
1uF
P3
P2
GND

CHIME SUPPRESS (K3, NC SSR)
opens the C1 path (P4<->CHIME_POS);
mutes gong; line 4 + latch intact

K3
GAQY412EH
MUTE_DRV
R6
220
K3_A
R9
10k
P4
CHIME_POS
GND

K4
GAQY412EH
DOOR_DRV
R21
220
K4_A
Pull-down for DOOR_DRV is on K2
SEAL_IN
K5_COM
GND

UNLATCH K5 (K4, NC SSR)

K6
GAQY412EH
P4_ISO
R12
10k
K5_SENSE_N
R34
220
K6_A
+3V3
R44
100k
R35
10k
K6_RET
JP2
K6_BYPASS
P4
K5_LATCH
GND

ISOLATE P4 (K6, NC SSR)

BELL SENSE

J2
WF26 bus
P1
P2
P3
P4
P5
D2
H24VND3BA
D3
H24VND3BA
D7
H24VND3BA
D12
H24VND3BA
HOUSE BELL / Türruf (P1->P4)

D8
1N4148W
R1
5.1k
OC1
TLP293 (GB)
OC1_K
R22
12k
P4_SENSE_N
+3V3
GND

D9
1N4148W
R2
5.1k
OC2
TLP293 (GB)
OC2_K
R23
12k
P5_SENSE_N
+3V3
GND

APARTMENT BELL / Etagenruf (P1->P5)

RESET / BOOT

+3V3
R10
10k
EN
C5
1uF
SW2
RST
GND

R11
10k
BOOT
SW1
BOOT
GND

AUDIO ES8311

+3V3
R19
10k
C7
100nF
C8
100nF
AVDD_PRE
D18
LMBR01S30ST5G
AVDD
C9
100nF
R37
220
C10
10uF
+5V +3V3
LP5907MFX-3.3
U4
IN
OUT
EN
GND
C23
10uF
C24
10uF
AVDD_PRE
+3V3
R18
10k
I2C_SDA
I2C_SCL
P1
ES_CE
R20
10k
U3
ES8311
CCLK
MCLK
PVDD
DVDD
DGND
SCLK/DMIC_SCL
ASDOUT
LRCK
DSDIN
AGND
EP
CE
CDATA
MIC1P/DMIC_SDA
MIC1N
VMID
ADCVREF
DACVREF
OUTN
OUTP
AVDD
ES_VMID
I2S_MCLK
I2S_BCLK
I2S_DIN
I2S_WS
I2S_DOUT
D13
D16
LMBR01S30ST5G
ES_OUTP
R26
22k
ES_OUTP_AC
C14
1uF
ES_OUTN
R16
22k
ES_OUTN_AC
C15
1uF
ES_MICN
R31
22k
ES_MICN_AC
C17
1uF
D14
D17
LMBR01S30ST5G
ES_MICP
R30
22k
ES_MICP_AC
C16
1uF
ES_DACVREF
C11
10uF
ES_ADCVREF
C13
1uF
ES_MICN
R32
3.3k
ES_MICP
R33
3.3k
ES_VMID
C12
10uF
TALK_BRIDGE
P2
GND

TEST POINTS

TP1
GND
+3V3
TP2
TP3
WD_GATE
TP4
AVDD_PRE
TP5
ES_OUTP
TP6
ES_MICP
TP7
ES_MICN
TP8
ES_VMID
TP9
TALK_BRIDGE
TP10
K5_LATCH
TP11
AVDD

WF 26/G

SW3
Tueroeffner (door release)
MP2
P3
NO2
NO1
P2
COM2
COM1
SEAL_IN
NC2
NC1
MP1
GND
K5_LATCH
K5_LATCH
K5_LATCH
K6_RET
D1
1N4004W
K5
G6K-2F-Y DC12
P1
K5_COM
GND

CHIME_POS
R36
100k
CHIME_BLEED_RET
C19
22uF/50V
JP1
CHIME_BLEED
P5
GND
K5_LATCH
R29
2.2k
TALK_SW
SW4
Sprechen/Hoeren (talk)
PTT_SW_N
K1_LED_RET
R43
1k
+3V3
R42
10k
PTT_SENSE_N
GND
LS1
P1
Speaker/Mic 16R

H1
MountingHole
H2
MountingHole

Gunnar Beutner
Sheet: /
File: doorbell.kicad_sch
Title: Doorbell controller
Size: A3
Date: 2026-07-02
Rev: V4.2
KiCad E.D.A. 10.0.4
Id: 1/1